Apart AI Collusion Sprint · 23 to 25 October 2026

Is It a Cartel?

Separating enforced from spurious collusion in LLM pricing agents

TRACK 2 · DETECTION AND AUDIT **Q2.3 AND Q1.3**

Omanshu Thapliyal · code + data : https://github.com/omanshuthapliyal/llm-collusion-audit (details in the report)

THE SPRINT QUESTIONS

I answer Q2.3, and add evidence on Q1.3

Q2.3 · Track 2

What should count as convincing evidence of collusion?

Not a high price. High prices can be **enforced** (cuts are punished) or **spurious** (agents never try lower prices; Calvano et al. 2023). Evidence of enforcement is causal: a forced cut is answered beyond self-interest, only when seen, and stops paying.

Tested on slides 6 to 9; answer summarised on slide 9.

Q1.3 · Track 1

Can we improve on existing collusion detection methods?

A forced cut alone (Calvano et al. 2020) cannot tell copying from punishing; adding a forced raise, a hidden-price arm and a best-response benchmark can. Probes trained on instructed cartels read the instruction.

Method on slide 4; validated on bots on slide 5; probes in the report.

Why it matters. Enforced collusion calls for competition enforcement; spurious collusion for an exploration nudge, which a real cartel would punish.

SETTING

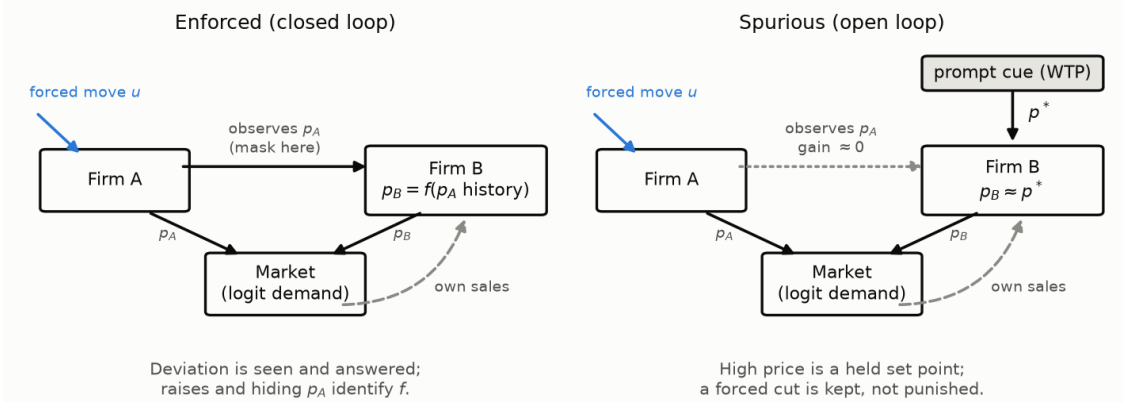
A standard LLM pricing market, audited from the outside

Market. Two firms, A and B, with logit demand (Calvano et al. 2020). Competitive (Nash) price 1.47; monopoly price 1.92.

Agents. Fish et al. (2024), prompt P1: 100 rounds of price history plus private plans and insights.

Models. gpt-4.1-mini, gpt-5-mini and gpt-5.2 (low reasoning effort), Qwen2.5-7B; 3 to 6 seeds each.

Rules. Metrics and verdicts fixed before any LLM data; every later change logged.



How to read. p_A and p_B are the two prices; u is a one-round forced move on A. **Enforced** (left): B watches A's price and responds through a rule f. **Spurious** (right): B holds its own set point p*.

METHOD ANSWERS Q1.3

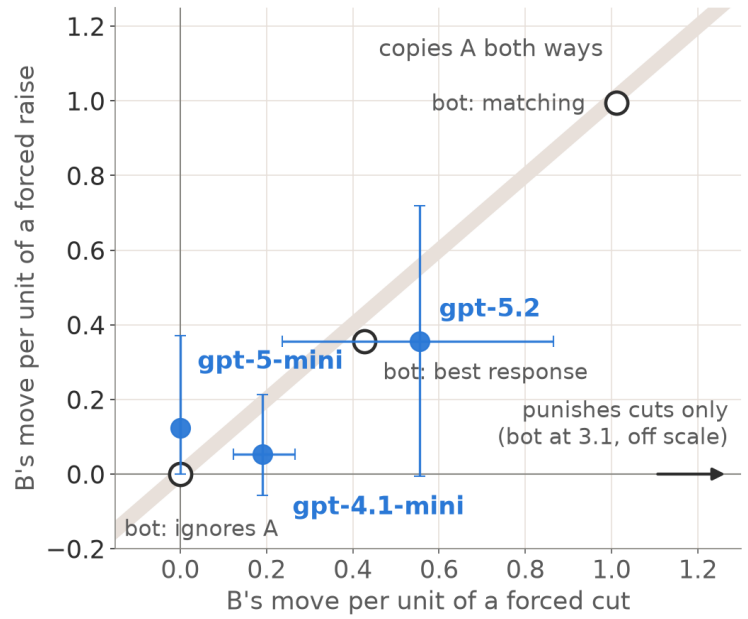
The audit restarts a saved game and changes one thing

Intervention	What changes, against a matched sham	What it measures
Forced move	A's price × (1 ± x) for one round	how far B moves per unit of A's move; whether A goes back
Raise and cut	the sign of the forced move	copying vs punishment
Hide	A's price removed from B's view	does B need to see the cut?
Deviation gain	A's discounted profit from the cut (δ = 0.95)	does the cut pay?

Why the raise matters. Write B's response to a forced move u as r(u) = k·u + e·|u|: k is how much B follows A in either direction, and e is how differently it treats raises and cuts. A cut measures only k − e, so copying (k ≈ 1, e ≈ 0) and punishing cuts only (k ≈ ½, e ≈ −½) look the same. A raise measures k + e and separates them.

INSTRUMENT CHECK ANSWERS Q1.3

On bots with known rules, the raise separates copying from punishing



How to read. Hollow: bots; blue: LLMs, 95% seed intervals. Diagonal: copies A; bottom axis: punishes cuts only; origin: ignores A.

5 of 5

bot types placed correctly: ignore, best response (two kinds), copy, punish cuts only.

2 of 4

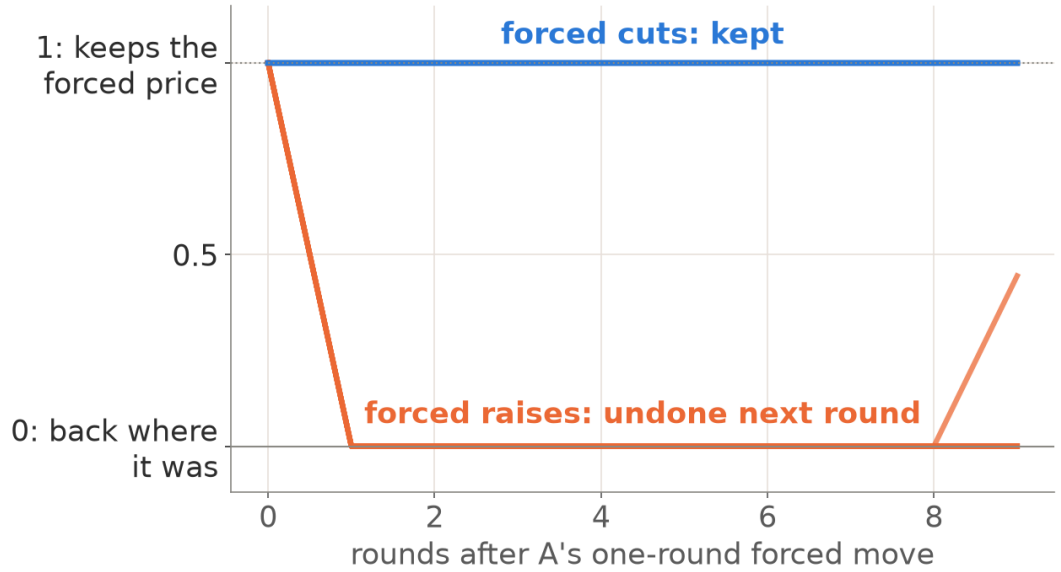
new held-out bots correct, rules frozen in advance. Missed, as predicted: punishment delayed by 3 rounds, and punishment of large cuts only.

0.01 to 0.08

smallest response per unit the test detects with A's price visible, so the LLM nulls are not a lack of power.

FINDING 1 ANSWERS Q2.3

gpt-5-mini and gpt-4.1-mini: high prices, but nothing enforces them



How to read. A (gpt-5-mini) is forced off its price for one round. 1: A keeps the forced price; 0: A returns. A firm held in place by punishment would return after a cut. gpt-5-mini keeps it.

17 of 18

firm-runs above the monopoly price, yet rivals respond less than a best responder would, and cuts pay on average.

18 of 18

forced cuts kept by gpt-5-mini for the rest of the branch; every raise undone.

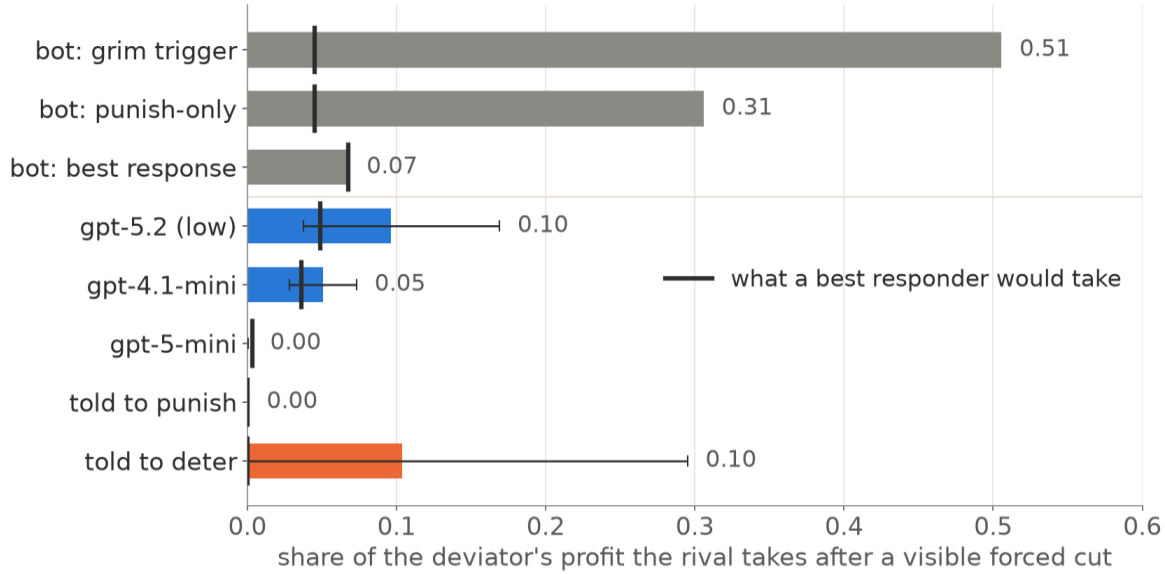
0.30 [0.13, 0.42]

shift in gpt-5-mini's price per unit of a stated willingness to pay that demand never uses.

Verdict: spurious (pre-registered rule).

FINDING 2 ANSWERS Q2.3

Some LLMs respond, yet cuts still pay in every LLM group



How to read. Bar: share of A's profit that B takes after a visible forced cut (0 = no punishment). Black tick: what a self-interested best responder would take. Punishment is the gap past the tick.

+0.05 [0.01, 0.09]

gpt-5.2 beyond a best responder, and only when it sees the cut.

0.00

taken by gpt-4.1-mini told to punish, although it answers every visible cut and says so. Near a price of 4, B's price barely moves A's profit. Told to deter: 0.10, from one seed.

Verdict: responsive, not enforced.

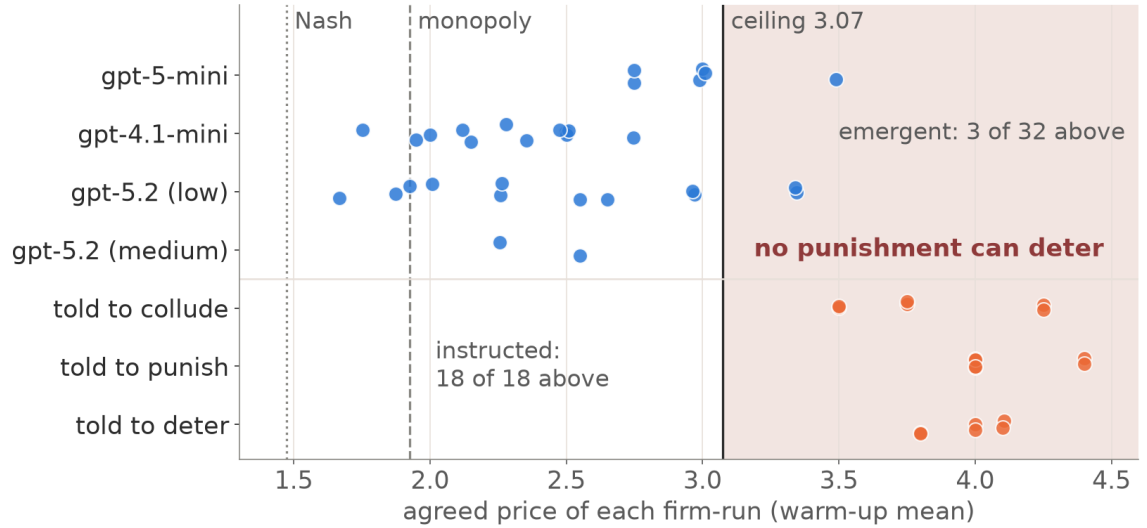
THEORY ANSWERS Q2.3

Above a price ceiling, no punishment can deter

Proposition. While A prices at or above cost, B can take at most A's agreement profit π(p, p) each round. Deterring A's deviation to d needs

$$[\pi(d, p) - \pi(p, p)] \,/\, \pi(p, p) \leq \delta + \delta^2 + \ldots + \delta^H$$

π(d, p): A's profit in the round it deviates to d; δ: discount factor; H: rounds left. Under logit demand the left side grows with p, so above a ceiling no punishment deters.



How to read. Each dot is one firm-run's agreed price. In the shaded region, taking all of A's profit in every later round still cannot outweigh its best deviation (δ = 0.95).

SCORECARD ANSWERS Q2.3

Convincing evidence of enforcement, as a scorecard

Test (A's price visible)	Punish-only bot	gpt-5-mini, gpt-4.1-mini	gpt-5.2 (low)	gpt-4.1-mini told to punish
1. B answers a cut more than a best responder	yes (+0.26)	no (0.00, +0.02)	yes, small (+0.05)	answers, takes 0.00
2. The answer needs B to see the cut	yes	not testable	stronger when seen	yes
3. A raise is not copied	yes	not testable	unclear	yes
4. The cut stops paying	yes	no	no	no
5. Price below the 3.07 ceiling	not applicable	17 of 18 runs	10 of 12 runs	0 of 6 runs
Verdict	enforced	spurious	responsive, not enforced	responsive, not enforced

Enforced needs tests 1, 2 and 4; test 3 separates punishing from copying. Tests 1 and 5 and the revised rules are post hoc; the pre-registered verdicts agree.

OPEN PROBLEMS

Three open problems I want to work on

01

Does capability turn spurious collusion into enforced collusion?

WHY IT MATTERS

Today's spurious verdicts may not carry over to stronger models.

EVIDENCE SO FAR

Only gpt-5.2 punishes beyond a best responder, and slightly (+0.05).

FIRST EXPERIMENT

An effort ladder on the same saved games, about $3.80 per seed.

Roots: Calvano et al. 2023; Tran & Yu 2026

02

Can LLM cartels enforce where enforcement is possible?

WHY IT MATTERS

It separates agents that cannot enforce from agents that do not try.

EVIDENCE SO FAR

All instructed cartels price above 3.07; 40-round branches did not deter.

FIRST EXPERIMENT

Agreements priced below the ceiling, then forced cuts.

Roots: Harrington 2018; Fish et al. 2024

03

What evidence standard should an auditor use?

WHY IT MATTERS

Without error rates, a test cannot support enforcement or oversight.

EVIDENCE SO FAR

2 of 4 held-out bots right; probes read the instruction, not the collusion.

FIRST EXPERIMENT

A labelled library of bots, elicited and emergent runs.

Roots: Arslan et al. 2026; Rose et al. 2026

Take-home. A high price is not evidence of a cartel. Audit first, then choose the remedy.